

Implement Backup & Recovery

Recovery Services Vault, Azure Backup policies, VM backup and restore, Azure Site Recovery, soft delete, and the difference between backup, DR, and snapshots — everything you need for the final exam domain.

Two vaults, two jobs — know which one does what

Azure backup and recovery centers on two vault types. The Recovery Services Vault handles backup and site recovery for on-premises and Azure workloads. The Backup Vault is newer, optimized for Azure-native data sources like disks and blobs. The exam tests both — know the difference.

RECOVERY SERVICES VAULT

Backup + Site Recovery

- Azure VM backup (IaaS)
- SQL Server in Azure VMs
- Azure Files backup
- On-premises backup (MARS agent)
- Azure Site Recovery (DR replication)
- Supports LRS, ZRS, GRS redundancy

BACKUP VAULT

Azure-native data sources

- Azure Disk backup (managed disks)
- Azure Blob backup (operational)
- Azure Database for PostgreSQL
- AKS cluster backup
- Newer vault type — expanding support
- Supports LRS, ZRS redundancy

Exam rule: VM backup → Recovery Services Vault. Managed disk backup → Backup Vault. Azure Files backup → Recovery Services Vault. If in doubt, Recovery Services Vault is the answer for most AZ-104 backup scenarios.

Eight terms you need cold

Backup Policy

Defines the backup schedule (daily/weekly), retention (daily, weekly, monthly, yearly), and consistency type. Each vault item is associated with exactly one policy.

Tip: You can change the policy assigned to a backup item at any time — existing recovery points are retained under the old policy until they expire.

Recovery Point

A snapshot of a resource at a specific point in time stored in the vault. Restoring from a recovery point returns the resource to that exact state.

Tip: VM recovery points can be Application Consistent (VSS snapshot — ensures app-level consistency), File System Consistent, or Crash Consistent (lowest overhead).

Soft Delete

A vault-level setting that retains deleted backup data for 14 additional days (free) before permanent deletion — protects against accidental or malicious backup deletion.

Tip: Soft delete is enabled by default on new Recovery Services Vaults. To permanently delete a backup item, you must first disable soft delete or wait for the retention period.

Azure Site Recovery (ASR)

A Disaster Recovery as a Service (DRaaS) solution that continuously replicates VMs and physical servers to a secondary region — enables failover and failback with defined RPO/RTO.

Tip: ASR is also used for VM migration between regions and from on-premises to Azure — not just for DR.

RPO / RTO

RPO (Recovery Point Objective) — the maximum acceptable data loss measured in time. **RTO** (Recovery Time Objective) — the maximum acceptable downtime before recovery must complete.

Tip: Backup meets higher RPO/RTO requirements (hours). ASR continuous replication targets near-zero RPO and RTO in minutes.

MARS Agent

Microsoft Azure Recovery Services agent — installed on Windows machines (physical or VM) to back up files, folders, and system state directly to a Recovery Services Vault.

Tip: MARS agent backs up files/folders only — not full disk images. For full VM backup, use the Azure Backup VM extension instead.

Vault Redundancy

How the vault stores backup data: LRS (3 copies in one datacenter), ZRS (3 copies across zones), GRS (6 copies across two regions). GRS is the default and enables Cross-Region Restore.

Tip: Vault redundancy can only be changed before the first backup item is protected — it cannot be changed after data is stored in the vault.

> Cross-Region Restore

Allows restoring a VM backup to the secondary GRS region — useful when the primary region is unavailable. Only available when the vault uses GRS redundancy.

Tip: Cross-Region Restore must be explicitly enabled on the vault — GRS alone is not enough to allow secondary region restores.

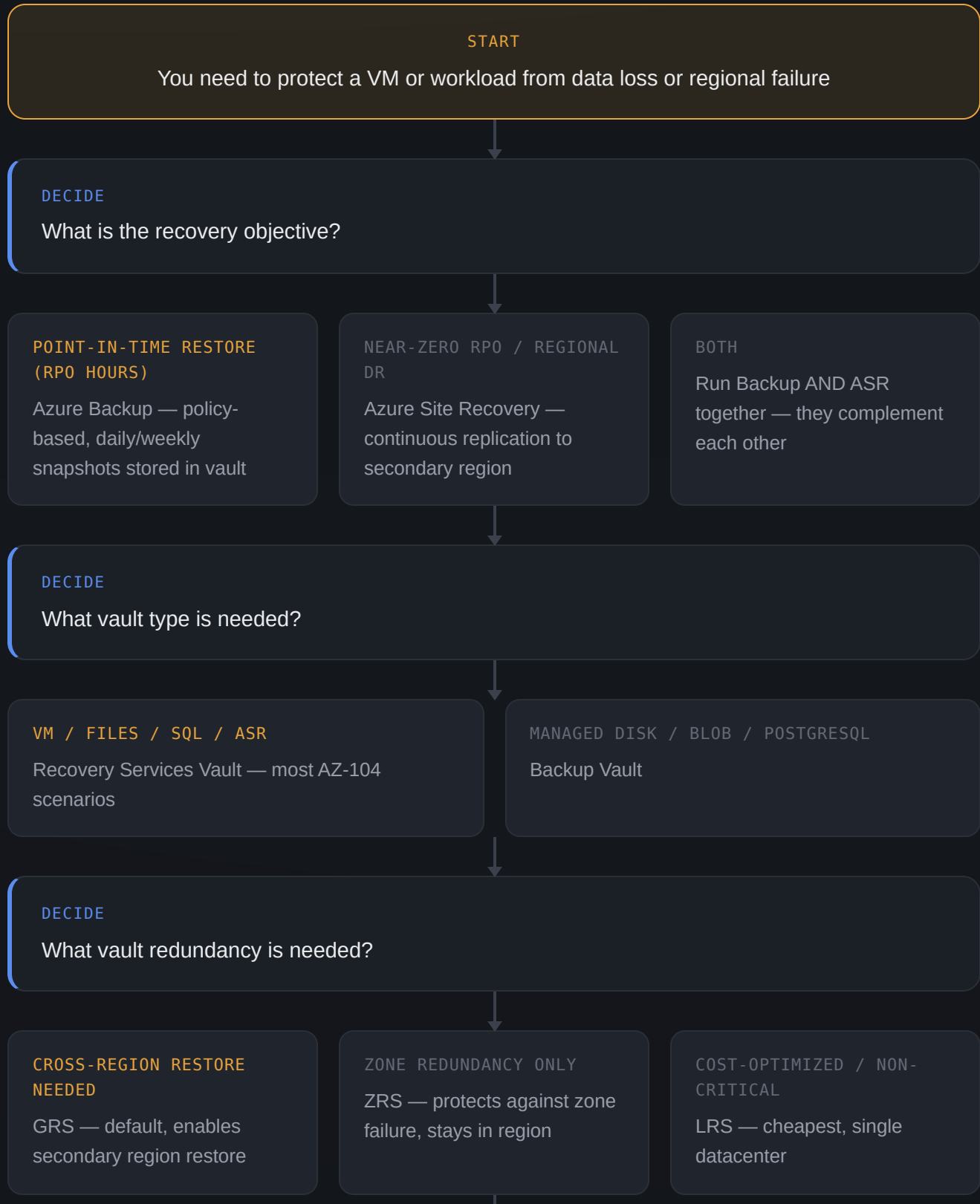
WORKLOAD	SERVICE	VAULT TYPE	AGENT NEEDED?
Azure VM (full disk)	Azure Backup	Recovery Services Vault	No — extension auto-installed
Azure Files share	Azure Backup	Recovery Services Vault	No
SQL Server in Azure VM	Azure Backup	Recovery Services Vault	Yes — SQL backup extension
Managed Disk backup	Azure Backup	Backup Vault	No
On-premises files/folders	Azure Backup (MARS)	Recovery Services Vault	Yes — MARS agent
VM disaster recovery	Azure Site Recovery	Recovery Services Vault	Yes — Mobility service

EXAM GOTCHA — BACKUP VS SNAPSHOTS VS ASR

Snapshots live in the same storage account as the disk — they don't protect against account-level deletion or regional failure. **Azure Backup** stores data in a vault in a separate storage domain with its own redundancy — protects against accidental deletion (soft delete) and supports cross-region restore. **Azure Site Recovery** continuously replicates to a separate region for near-zero RPO DR — it is not a backup solution and cannot restore to a historical point in time.

Setting up backup and DR for a VM

Walk through this for every VM or workload that needs data protection.



CONFIGURE

Create a Backup Policy — set schedule (daily/weekly) and retention (daily, weekly, monthly, yearly GFS)

PROTECT

Enable backup on each VM — Azure Backup auto-installs the VM extension on first backup

DECIDE

Should backup data be protected from accidental deletion?

YES (ALWAYS RECOMMENDED)

Confirm soft delete is enabled on the vault — it is on by default for new vaults

NO

Disable soft delete — not recommended for production

DONE

VM protected with backup policy, vault redundancy set, soft delete active

Tools for backup and recovery

>_ Azure CLI — Vault & Policy

BEST FOR: scripted vault creation and policy setup

Create vaults, set redundancy, and create backup policies in a deployment script — ensure every environment has backup configured at deployment time.

```
az backup vault create \
  -g rg-prod -n rsv-prod \
  --location eastus

az backup vault backup-properties
set \
  -g rg-prod -n rsv-prod \
  --backup-storage-redundancy
GeoRedundant

az backup policy set \
  -g rg-prod -v rsv-prod \
  --policy @vm-backup-policy.json \
  --name policy-vm-daily
```

>_ Azure CLI — Enable & Trigger Backup

BEST FOR: enabling backup on VMs and triggering on-demand jobs

Enable backup on a VM and trigger an immediate on-demand backup — useful for pre-maintenance safety snapshots.

```
az backup protection enable-for-vm \
-g rg-prod -v rsv-prod \ --vm
vm-web01 \ --policy-name policy-vm-daily # Trigger an on-demand
backup now az backup protection
backup-now \ -g rg-prod -v rsv-prod \ --container-name vm-
web01 \ --item-name vm-web01 \
--backup-management-type
AzureIaaSVM \ --retain-until 2026-
12-31
```

> Azure CLI — List & Restore

BEST FOR: listing recovery points and triggering restores

List available recovery points and trigger a restore to original location or a new VM — the key commands for recovery operations.

```
az backup recoverypoint list \-g rg-prod -v rsv-prod \--container-name vm-web01 \--item-name vm-web01 \--backup-management-type AzureIaaSVM \--workload-type VM --output table
az backup restore restore-disks \-g rg-prod -v rsv-prod \--container-name vm-web01 \--item-name vm-web01 \--rp-name <recovery-point-name> \--storage-account mystorageacct \--target-resource-group rg-restore
```

<> Bicep — Vault + Backup Policy

BEST FOR: IaC — backup as part of VM deployment

Deploy the vault and backup policy alongside VMs — protection is configured at deployment, never forgotten.

```
resource vault
'Microsoft.RecoveryServices/vaults@2023-04-01' = {
  name: 'rsv-prod'
  location: location
  sku: { name: 'RS0', tier: 'Standard' }
  properties: {}
}

resource backupPolicy
'Microsoft.RecoveryServices/vaults/backupPolicies@2023-04-01' = {
  name: 'policy-vm-daily'
  parent: vault
  properties: {
    backupManagementType: 'AzureIaaSVM'
    schedulePolicy: {
      schedulePolicyType: 'SimpleSchedulePolicy'
      scheduleRunFrequency: 'Daily'
      scheduleRunTimes: ['2026-01-01T02:00:00Z']
    }
    retentionPolicy: {
      retentionPolicyType: 'LongTermRetentionPolicy'
      dailySchedule: {
        retentionDuration: { count: 30, durationType: 'Days' }
      }
    }
  }
}
```


Portal — ASR Setup

BEST FOR: enabling Site Recovery replication

The ASR wizard in the portal walks through source region, target region, replication policy, and network mapping — the most guided way to set up DR replication for VMs.

Recovery Services Vault

- Site Recovery → + Replicate
- Source: Azure → Source region:

East US

- Target region: West US
- Select VMs → Configure settings
- Enable replication

Test failover (no production impact)

- Replicated items → VM → Test failover
- Select recovery point → Run test

RESTORE OPTION	WHAT IT DOES	WHEN TO USE
Restore VM	Creates a new VM from a recovery point	Full VM recovery to a new machine
Restore Disks	Restores managed disks to a storage account — attach manually	Selective disk recovery or cross-subscription restore
Replace Existing Disk	Replaces the OS or data disk on the existing VM in place	Corrupted disk on a running VM with fixed config
File Recovery	Mounts a recovery point as a drive — browse and copy individual files	Recover specific files without restoring the whole VM
Cross-Region Restore	Restores to the GRS secondary region	Primary region outage — requires GRS vault + CRR enabled

EXAM GOTCHA – VAULT REDUNDANCY IS LOCKED AFTER FIRST BACKUP

The vault's storage redundancy (LRS / ZRS / GRS) can **only be changed before the first backup item is protected**. Once any VM or file share is backed up to the vault, the redundancy setting is locked permanently. Plan redundancy before enabling protection — not after.

Principles worth internalizing

PROTECTION TYPE	RPO	RTO	POINT-IN-TIME RESTORE?	CROSS-REGION?
Azure Backup (daily)	Up to 24 hours	Hours	Yes	Yes — with GRS + CRR
Azure Backup (hourly)	Up to 4 hours	Hours	Yes	Yes — with GRS + CRR
Azure Site Recovery	~30 seconds	Minutes	No — latest replicated state only	Yes — core purpose
VM Snapshot (manual)	Point of snapshot	Minutes	Yes	No — same account

SOFT DELETE – THE EXAM'S FAVORITE PROTECTION FEATURE

Soft delete is **enabled by default** on new Recovery Services Vaults. When a backup item is deleted, the data is retained for **14 extra days at no charge** — marked as "soft deleted" and can be undeleted. To permanently remove data, you must explicitly disable soft delete (which takes effect immediately) or wait 14 days. The exam tests: what happens when you delete a VM but soft delete is on, and how to force-delete backup data before the 14-day window expires.

Six mistakes that show up on the exam

Treating ASR as a backup solution

ASR replicates to the latest state — it cannot restore to a specific point in time. Use Azure Backup for point-in-time recovery; use ASR for regional DR failover.

Changing vault redundancy after enabling backup

Vault redundancy is locked after the first backup item is protected. Set GRS before enabling any backup — you cannot change it later without starting over with a new vault.

Enabling Cross-Region Restore without setting GRS first

CRR requires GRS vault redundancy — it must also be explicitly enabled on the vault even when GRS is set. Both are required; neither alone is sufficient.

Confusing MARS agent with the VM backup extension

MARS backs up files and folders from Windows machines. The Azure Backup VM extension backs up the full disk image. They're different agents for different purposes.

Relying on soft delete to be disabled by default on old vaults

Soft delete is enabled by default on new vaults but may not be on vaults created before the feature rolled out. Always verify and explicitly enable it in production.

Using File Recovery when full VM restore is faster

File Recovery mounts the recovery point as a drive — useful for a few files but not for large restores. For recovering more than a handful of files, restoring the full VM or disk is usually faster.